Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Oslakovic, Keith E.		
Project Title:	Functionally-Adapted Biomimetic Materials for Space Exploration		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	oslakovic@mole-geodesics.com	**Fax:**	FY 617-427-1234
PI Organization Type:	INDUSTRY	**Phone:**	617-427-0300
Organization Name:	Molecular Geodesics, Inc.		
PI Address 1:	Engineering Manager		
PI Address 2:	20 Hampton Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02119	**Congressional District:**	8
Comments:	HQ BITR		
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	01/01/1999	**End Date:**	01/01/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

The overall goal of this research is to create biomimetic materials that will revolutionize how the aerospace industry designs and manufactures spacecraft for future manned space exploration. Storage tanks that hold fluids, such as fuel propellant and water, account for the major portion of the volume of spacecraft and roving space vehicles and thus, offer the greatest opportunity for weight reduction. Molecular Geodesics, Inc. (MGI) will focus on designing and manufacturing a potential replacement for these storage tanks in the form of a porous biomimetic scaffold that mimics the trabecular microarchitecture of living cells and tissues. A spacecraft containing a porous biomimetic fuselage shell will have the potential of redistributing the weight of the fuel and structure, effectively creating a larger internal area for additional payload, passengers, and day-to-day operations. Thus, we will essentially use "fuel as structure" to reduce weight and afford space savings, two critical needs for long term manned space exploration. MGI will adapt its existing proprietary computer-based design and manufacturing technologies to create functionally adapted biomimetic scaffolds

Task Description:	that can hold fluids, such as fuel propellant, while providing critical load-bearing and thermal insulation functions required in a fuselage shell. The specific tasks include 1) computer-assisted design of three-dimensional prototypes of porous biomimetic scaffolds with regular and functionally adapted microarchitecture, 2) computer-assisted manufacturing of prototypes of biomimetic trabecular fuselage shell components, and 3) mechanical testing of these prototypes. We will also identify key process design issues involved with scaling up this design and manufacturing technology for commercial production of larger scale biomimetic scaffold components and provide an assessment of the estimated manufacturing costs. We will essentially use "fuel as structure" to reduce weight and afford space savings, two critical needs for long term manned space exploration. The overall goal of this research is to create biomimetic materials that will revolutionize how the aerospace industry designs and manufactures spacecraft for future manned space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)